Course Title	Electric Vehicle System Engineering			Course Type	PE-2
Course Code	B24EES502	Credit	3	Class	V Semester
CI Hours / Sem		LI Hours/Sem	TW, SL in Hours / Sem	Total Hours / Sem	Total Credits
L	T	P	SL		
42	0	0	48	90	3

COURSE OVERVIEW

Electric Vehicle System Engineering program at REVA University, a uniquely industry-driven course, fully immersing students in hands-on learning experiences. With an academic MATLAB license (No. 41185143) at their disposal, students will undertake initial simulations in our cutting-edge Simulation Lab, designed to provide a practical foundation in electric vehicle systems. What sets our program apart is the collaboration with Bosch Pvt Ltd, where industry experts will actively engage with our students, bridging the divide between academia and real-world application. These experts will not on

enrich the learning experience but also guide students in successfully completing their course assignments, ensuring that they are well-prepared for the challenges and opportunities in the electric vehicle industry.

COURSE OBJECTIVE

1. To present a comprehensive overview of Electric Vehicles
2. To enable students to know the characteristics of various types of energy storage devices .
3. To present a comprehensive overview of Power Electronics and Electric Drive Train Systems.
4. To equip students with basic concepts to practically design/ implement and modify the existing vehicle to electric vehicle

COURSE OUTCOMES (COs)

After the completion of the course, the student will be able to:

CO#	Course Outcomes	POs	PSOs
CO-1	Describe the configuration of a typical electric vehicle, and design , develop basic a schemes of electric vehicle	1-6	1,2
CO-2	Choose proper energy storage systems for vehicle applications and develop strategies for charging various types of batteries.	1-3,6	1,2
CO-3	Understand the Power Electronics and Electric Drive Train Systems quadrant wise.	1,3,4	1,2
CO-4	Execute the Modeling of Vehicle Control Strategy	1,3,4	1,2

BLOOM'S LEVEL OF THE COURSE OUTCOMES

CO#	Bloom's Level					
	Remembe r (L1)	Understan d (L2)	Appl y (L3)	Analyze (L4)	Evaluate (L5)	Create (L6)
CO-1		√				
CO-2			√	√		
CO-3		√				
CO-4		√	√			√

COURSE ARTICULATION MATRIX

CO#/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO-1	2	1		1								2	2	
CO-2	3	1	2			1						2	2	
CO-3	3		2	1								2	2	
CO-4	3		2	1								2	2	

Note: 1-Low, 2-Medium, 3-High

COURSE CONTENTS THEORY

Unit 1: Introduction to Electric Vehicles

Introduction to EVs, Configuration of EVs, Performance of EVs, Energy Flow Analysis, Introduction to Traction Batteries. Drive cycle and its importance in system analysis. Component selection for a given drive cycle for a BEV. Three Phase Inverter, VSI, PWM Techniques- Space Vector PWM- Hysteresis Control, Comparison of PWM techniques

Module 2: Energy Storage System

Ragone plot, Battery Types and battery pack, Basic battery operation, Battery Chemistry -Li Ion, Electric Vehicle Battery Efficiency and Effects on Performance, Calculation of Electric Vehicle Battery Capacity, SoC and SoH estimation, Governing equations and Waveforms, Battery parameters and comparisons, BEV – battery sizing, Modelling of Battery

-RC equivalent network Model, Concept of cell balancing, Basic battery management systems. Battery Chargers- Basic requirements for charging systems, Classification of Charging Architectures, Charging Controls, Current Regulations, charging standards and technologies.

Unit 3: Introduction to Power Electronics and Electric Drive Train Systems

Power Electronic Converters for Electric and Hybrid Vehicles, Electric Vehicle Charger Topologies, Electric Drive train Overview, Systems with Linear Motion and Rotating Systems, Types of loads, Four Quadrant Operation, Induction Motor for EV power train - Steady State Analysis of Induction Drive, Direct & Indirect Vector Control, Field Oriented Control, Machine Model with Leakage Inductance. Permanent magnet synchronous machine for EV power train, Non-Salient & Salient Drives, Generic Model, Steady State Analysis, Field Oriented Control.

Unit 4: EV Modelling & Simulation

Modeling of BEV-Forward looking Model-Driver Perspective, Backward Looking Model Drive Cycle Perspective, Modelling of Driver, Modeling of Brake Control Unit/Regenerative braking , Modeling of Vehicle Control Strategy, Modeling of Vehicle Chassis Sizing of Components- Steady State Energy Balance Equation, Modeling of Battery/Cell, Powertrain Dimensioning Peak vs Continuous performance, Type of Drive cycles, Types of Control Strategy, Analysis Performance, Range, Consumption Prediction.

TEXT BOOKS:

1. C.C. Chan and K.T. Chau, Modern Electric Vehicle Technology, London: Oxford University Press
2. Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals, New York: RC Press.
3. M. Ehsani, Y. Gao, S .E. Gay and A. Emadi, Modern Electric, Hybrid Electric, and Fuel Cell Vehicles:

Reference Books:

1. Electric Vehicle Technology Explained, 2e, James Larminie, Oxford Brooks University John Lowry, Acenti Designs Ltd John Wiley & Sons, Inc., 2012, ISBN: 978-1-119-94273-3
2. Analysis and Control of Electric Drives: Simulations and Laboratory Implementation, Ned Mohan, University of Minnesota, Siddharth Raju, University of Minnesota, John Wiley & Sons, Inc., 2020 ISBN:

9781119584537
3. Electric Motor Control: DC, AC, and BLDC Motors, Sang-Hoon Kim, Kangwon National University Elsevier Science, 2017ISBN: 9780128121382
4. Renewable Energy Systems: Simulation with Simulink and Simscape Electrical Viktor Perelmuter, National Technical University (Kharkov Polytechnic Institute) CRC Press, Inc., 2017 ISBN: 978-1- 4987-6598-5
5. Modern Electric, Hybrid Electric, and Fuel Cell Vehicles, 3e Mehrdad Ehsani, Texas A&M University; Yimin Gao, Texas A&M University; Stefano Longo, Cranfield University; Kambiz M Ebrahimi, Loughborough University CRC Press, Inc., 2018 ISBN: 9781498761772
6. Battery Management Systems, Volume II: Equivalent-Circuit Methods, Gregory L. Plett, Publisher: Artech House, Year: 2020 ISBN: 1630810274,9781630810276
7. Fundamentals of Power Electronics, Robert W Erickson, Dragon Maksimovic, Springer, ISBN-10:8181283635, ISBN-13: 978-8181283634
8. Power Electronics: Converters, Applications and Design, Ned Mohan, Tore M. Undeland, and William P. Robbins , Wiley, ISBN-10: 9788126510900, ISBN-13: 978-8126510900

Journals/Magazines

1. www.myev.com
2. www.batteryuniversity.com
3. www.insideevs.com

SWAYAM/NPTEL/MOOCs:

1. https://online-learning.tudelft.nl/courses/electric-cars-introduction/
2. https://www.mooc-list.com/course/hybrid-vehicles-edx
3. https://www.mooc-list.com/course/electric-cars-introduction-edx
4. https://www.coursera.org/learn/electric-vehicles-mobility
5. https://www.udemy.com/course/electric-vehicle-certificate-course/